Assignment 5: Database Creation and Table Implementation

ProjectRapido – Bike & Cab Booking System
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ProgrammeBCA Semester 3 (2025)
CourseDatabase Management Systems
File nameAssignment_5_Database_Creation.pdf

1. Objective

Create the Rapido database and implement every table of the finalized design using SQL DDL. Relationships are declared with primary keys and foreign keys so the schema is ready for integrity tests (Assignment 6) and data load (Assignment 7).

2. Mapping from the ER design

ER entity / relationship	SQL table	Cardinality in DDL
USER	users	1 side of rides
VEHICLE	vehicles	1 side of captains
CAPTAIN	captains	M side of vehicles; 1 side of rides
RIDE	rides	M side of users and captains
PAYMENT	payments	1:1 with rides via UNIQUE ride_id
ZONE	zones	1 side of captain_zones

OPERATES IN (M:N)	captain_zones	composite primary key

Tables are created in dependency order: users and vehicles first, then captains, zones, rides, payments, captain_zones.

3. DDL script (MySQL)

```
-- Rapido Bike & Cab Booking System -- Finalized schema (MySQL 8+) used in Assignments 5-9 -- Student: Tamajong Bless Ntumnyoh | USN: 2511021240069 DROP DATABASE IF EXISTS rapido_db; CREATE DATABASE rapido_db; USE rapido_db; CREATE TABLE users ( user_id INT PRIMARY KEY AUTO_INCREMENT, full_name VARCHAR(100) NOT NULL, phone_number VARCHAR(15) NOT NULL UNIQUE, email VARCHAR(120) UNIQUE, created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ); CREATE TABLE vehicles ( vehicle_id INT PRIMARY KEY AUTO_INCREMENT, vehicle_type VARCHAR(20) NOT NULL, registration_number VARCHAR(20) NOT NULL UNIQUE, model VARCHAR(60) NOT NULL, capacity INT NOT NULL DEFAULT 1, CONSTRAINT chk_vehicle_type CHECK (vehicle_type IN ('Bike', 'Auto', 'Cab')), CONSTRAINT chk_capacity CHECK (capacity BETWEEN 1 AND 6) ); CREATE TABLE captains ( captain_id INT PRIMARY KEY AUTO_INCREMENT, full_name VARCHAR(100) NOT NULL, phone_number VARCHAR(15) NOT NULL UNIQUE, vehicle_id INT NOT NULL, rating DECIMAL(3,2) NOT NULL DEFAULT 0.00, CONSTRAINT fk_captain_vehicle FOREIGN KEY (vehicle_id) REFERENCES vehicles(vehicle_id) ON UPDATE CASCADE ON DELETE RESTRICT, CONSTRAINT chk_rating CHECK (rating BETWEEN 0.00 AND 5.00) ); CREATE TABLE rides ( ride_id INT PRIMARY KEY AUTO_INCREMENT, user_id INT NOT NULL, captain_id INT NOT NULL, pickup_location VARCHAR(120) NOT NULL, drop_location VARCHAR(120) NOT NULL, fare DECIMAL(8,2) NOT NULL, status VARCHAR(20) NOT NULL DEFAULT 'Requested', requested_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP, CONSTRAINT fk_ride_user FOREIGN KEY (user_id) REFERENCES users(user_id) ON UPDATE CASCADE ON DELETE RESTRICT, CONSTRAINT fk_ride_captain FOREIGN KEY (captain_id) REFERENCES captains(captain_id) ON UPDATE CASCADE ON DELETE RESTRICT, CONSTRAINT chk_fare CHECK (fare >= 0), CONSTRAINT chk_status CHECK (status IN ('Requested', 'Ongoing', 'Completed', 'Cancelled')) ); CREATE TABLE payments ( payment_id INT PRIMARY KEY AUTO_INCREMENT, ride_id INT NOT NULL UNIQUE, amount DECIMAL(8,2) NOT NULL, payment_method VARCHAR(20) NOT NULL, payment_status VARCHAR(20) NOT NULL DEFAULT 'Pending', payment_time DATETIME NULL, CONSTRAINT fk_payment_ride FOREIGN KEY (ride_id) REFERENCES rides(ride_id) ON UPDATE CASCADE ON DELETE RESTRICT, CONSTRAINT chk_amount CHECK (amount >= 0), CONSTRAINT chk_pay_method CHECK (payment_method IN ('UPI', 'Card', 'Cash', 'Wallet')), CONSTRAINT chk_pay_status CHECK (payment_status IN ('Pending', 'Success', 'Failed', 'Refunded')) ); CREATE TABLE zones ( zone_id INT PRIMARY KEY AUTO_INCREMENT, zone_name VARCHAR(80) NOT NULL, city VARCHAR(80) NOT NULL, CONSTRAINT uq_zone_city UNIQUE (zone_name, city) ); CREATE TABLE captain_zones ( captain_id INT NOT NULL, zone_id INT NOT NULL, PRIMARY KEY (captain_id, zone_id), CONSTRAINT fk_cz_captain FOREIGN KEY (captain_id) REFERENCES captains(captain_id) ON UPDATE CASCADE ON DELETE CASCADE, CONSTRAINT fk_cz_zone FOREIGN KEY (zone_id) REFERENCES zones(zone_id) ON UPDATE CASCADE ON DELETE CASCADE );
```

4. Verification that tables exist

After execution, the catalogue lists seven application tables:

table_name
captain_zones
captains
payments
rides
users
vehicles
zones

Column structure (equivalent DESCRIBE / PRAGMA table_info):

users

cid	name	type	notnull	dflt_value	pk
0	user_id	INTEGER	0	NULL	1
1	full_name	TEXT	1	NULL	0
2	phone_number	TEXT	1	NULL	0
3	email	TEXT	0	NULL	0
4	created_at	TEXT	1	CURRENT_TIMESTAMP	0

vehicles

cid	name	type	notnull	dflt_value	pk
0	vehicle_id	INTEGER	0	NULL	1
1	vehicle_type	TEXT	1	NULL	0
2	registration_number	TEXT	1	NULL	0
3	model	TEXT	1	NULL	0
4	capacity	INTEGER	1	1	0

captains

cid	name	type	notnull	dflt_value	pk
0	captain_id	INTEGER	0	NULL	1
1	full_name	TEXT	1	NULL	0
2	phone_number	TEXT	1	NULL	0
3	vehicle_id	INTEGER	1	NULL	0
4	rating	REAL	1	0.00	0

rides

cid	name	type	notnull	dflt_value	pk
0	ride_id	INTEGER	0	NULL	1
1	user_id	INTEGER	1	NULL	0
2	captain_id	INTEGER	1	NULL	0
3	pickup_location	TEXT	1	NULL	0
4	drop_location	TEXT	1	NULL	0
5	fare	REAL	1	NULL	0
6	status	TEXT	1	'Requested'	0
7	requested_at	TEXT	1	CURRENT_TIMESTAMP	0

payments

cid	name	type	notnull	dflt_value	pk
0	payment_id	INTEGER	0	NULL	1
1	ride_id	INTEGER	1	NULL	0
2	amount	REAL	1	NULL	0
3	payment_method	TEXT	1	NULL	0

cid	name	type	notnull	dflt_value	pk
4	payment_status	TEXT	1	'Pending'	0
5	payment_time	TEXT	0	NULL	0

zones

cid	name	type	notnull	dflt_value	pk
0	zone_id	INTEGER	0	NULL	1
1	zone_name	TEXT	1	NULL	0
2	city	TEXT	1	NULL	0

captain_zones

cid	name	type	notnull	dflt_value	pk
0	captain_id	INTEGER	1	NULL	1
1	zone_id	INTEGER	1	NULL	2

5. Relationships that were defined

Child table	FK column(s)	Parent	ON DELETE	ON UPDATE
captains	vehicle_id	vehicles.vehicle_id	RESTRICT	CASCADE

rides	user_id	users.user_id	RESTRICT	CASCADE
rides	captain_id	captains.captain_id	RESTRICT	CASCADE
payments	ride_id	rides.ride_id	RESTRICT	CASCADE
captain_zones	captain_id	captains.captain_id	CASCADE	CASCADE
captain_zones	zone_id	zones.zone_id	CASCADE	CASCADE

CREATE DATABASE / USE are shown for MySQL. The same tables were also executed in SQLite (INTEGER PRIMARY KEY AUTOINCREMENT) to capture the verification outputs included in these PDFs. Application logic is identical.